REQUEST FOR BIDS

15kV Switchgear for Substation 6

As per attached specifications

Sealed bids are to be addressed to:

Logan City
Attn: Lori Mathys
290 North 100 West
Logan, UT 84321-7806

Sealed bids will be accepted until 11:00 a.m., September 29, 2017.

The Purchasing Agent shall have possession of the bids at the designated time. Any bid en-route, either in the mail or at other locations in the city, will not be considered timely and will be returned unopened.

Bid package should be identified as "Sub 6 Switchgear" on the exterior envelope.

Questions regarding bid specifications should be directed to Mark Montgomery, Director, Logan Light & Power Department, at mark.montgomery@loganutah.org. All questions and clarifications must be received by 5:00 p.m., September 22, 2017.

The right is reserved by city to reject any or all bids.

Dated this September 8, 2017.

Lori Mathys
Purchasing Agent



Specification

15kV Switchgear

“Substation 6”

Logan City Utah
Logan, UT

09/08/2017

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GENERAL DESCRIPTION

1.1 PURPOSE OF SPECIFICATIONS

A. Specifications are intended to outline the general requirements, standards of performance, specific equipment characteristics, standards of materials and manufacturing, factory and/or on-site tests and warranty related to furnishing of equipment under these specifications.

B. Specifications provide the Manufacturer with minimum standards of quality for components and materials to be used in the manufacture of equipment. All materials and components shall be first quality, newly manufactured, from manufacture's or Manufacturer 's stock.

1.2 IDENTIFICATION OF PARTIES

A. "Owner":

Logan City Utah
530 North 800 West
Logan, UT 84321

Correspondence with the Owner shall be addressed to Mark Montgomery.

B. "Manufacturer ":

The firm, company, or corporation, with whom the Owner may select for the purpose of supplying switchgear described in these specifications.

1.3 WORK AND MATERIAL TO BE PROVIDED BY OTHERS

A. The following work and material will be provided by others unless specifically included in Manufacturer 's proposal. The Owner's bid evaluation will consider additional work or services offered by the Manufacturer .
   1. Termination of all field control and indication of cables.
   2. All required field assembly, including connection of overhead buswork and grounds.

1.4 METHOD OF BIDDING

A. Equivalent products:

1. When the words “or equal” are used in connection with the name of any item of equipment or material, or trade name they shall be construed to mean that the item named will be used as a basis of comparison and that all makes of similar items will be considered, providing they have equal or better qualifications. It is intended that products of equal quality and performance by other Manufacturer’s may be substituted.
2. Furnish drawings or other data as required to indicate all modifications resulting from use of such substitute items. Furnish general arrangement drawings, full descriptive data, and other information required to demonstrate to Engineer that material or equipment proposed is, in fact, equal to that specified. Burden of proof of equality shall in all cases remain with Manufacturer.
3. Abide by owner’s decision when proposed substitutions of material or equipment are deemed to be unacceptable.
4. Owner may consider such factors as over-all project arrangement, over-all project cost, and similar factors in determining acceptability of “equal” to that specified.

B. Form of bid price submittal:

1. Lump sum for all items as listed in the material Specification and Proposed Data Sheets.
2. Bid price shall cover complete work described in specifications, including costs incidental thereto, unless specifically indicated otherwise.
3. List separately the price of services of Service Engineer, as determined in the Proposed Data Sheets.

C. Bid submittals shall be in a sealed format to:

Logan City
Attn: Lori Mathys
290 North 100 West
Logan, UT 84321-7806

and are due not later than 11:00 a.m., September 29, 2017.

The desired switchgear delivery date is **as soon as possible. Delivery times may be a factor in selecting a bid.**

1.5 DATA TO ACCOMPANY BIDS

A. The complete bid document shall be properly filled out in every detail and submitted to the Owner, along with all other material as listed below:

1. Any exceptions taken by the Bidder to the Specifications, Proposed Data Sheets, and/or Terms and Conditions at time of bid, shall be clearly and simply stated or summarized, using an itemized format, in a specially prepared letter of transmittal attached to and made a part of the Proposed Data Sheets. NOTE: Manufacturer 's "Standard Conditions of Sale" catalog pages are not acceptable for purposes of stating exceptions to the specifications.

B. Manufacturer 's proposal shall also contain the following:

1. Manufacturer 's supplemental descriptive data on equipment proposed.
2. Itemized list of special tools which Bidder proposed to furnish.
3. Proposed arrangement drawings of the switchgear showing plan and elevation views with dimensions and weights for each Bid item. (Base and Alternates)
4. Standard and specified accessories, instruments, and spare parts to be furnished separately shall be listed in detail. List shall clearly define those items to be shipped separately. It is understood that all items not so listed shall be shipped, mounted and connected. Use proposed Data Sheets for listing.
5. Complete warranty, which the Manufacturer will offer, including option for extended guarantee through three (3) years' or five (5) years' total guarantee period, is requested.
6. Listing of known failures for switchgear of similar design and rating to those proposed, with time of failure (during test, field) and cause of failure, if known.

1.6 DELIVERY AND SHIPMENT

A. Bids shall include delivery F.O.B., freight prepaid, to Logan City project job site, Switchgear Pad, located at 1000 North 300 East (Substation 6), Logan Utah. The project site is located within the City limits of Logan, Utah. Delivery date shall be on or before that specified by the Manufacturer in the Proposal Data Sheets. Switchgear and all required components shall be delivered by means of a single shipment. The Manufacturer shall assume all responsibility for the safe delivery of equipment shipped F.O.B. SWITCHGEAR PAD. Upon request, Owner can provide local contractors with the equipment/expertise to unload Switchgear to pad.

B. Switchgear shall be shipped by motor freight as completely assembled as possible insofar as is consistent with good shipping practice. The equipment shall be carefully blocked and secured for shipment. If items must be

disassembled for shipment, they shall be tagged or stenciled with proper identification. A complete packing list shall also accompany each shipment.

C. Prior to shipment, the Manufacturer shall inform the Owner of the estimated time in shipment, routing, shipping company name and PRO number, if shipped by common carrier. Actual delivery time to be confirmed by telephone to Owner at least one (1) working day in advance.

1.7 DRAWINGS AND INSTRUCTION MANUALS

A. Manufacturing Progress Reports

1. Within 30 days of order, the Manufacturer shall furnish to the Owner a manufacturing schedule for switchgear to be furnished under this specification. Manufacturer ’s schedule shall include projected time allotment for engineering, approval drawing delivery, Owner’s drawing review period, fabrication drawing delivery, major materials delivery, all major divisions of fabrication and assembly, testing, and shipment. This schedule shall allow for work completed on each item to be shown in conjunction with projected schedule in a manner which allows updating for furnishing progress reports.
2. Progress reports shall be furnished to the Owner monthly, with each report showing progress from the previous month. However, the Owner may request progress reports on a more frequent basis should the Manufacturer ’s progress seriously fall behind his projected schedule.

B. Shop Drawings

“Shop Drawings” refers to all the detailed installation drawings prepared by the Manufacturer and/or his suppliers required to construct or install the switchgear. A general drawing format of ANSI Size B (11” X 17”) is preferred. The size of all drawings submitted as part of this project shall be a multiple of 8.5” X 11”. “Shop Drawings” shall include custom and non-universal drawings and data as follows:

1. Bill of materials with manufacturer, model/type, description, and catalog number for standard components and accessories. This list shall include circuit breakers, switches, insulators, components for operator mechanisms, relays, indicators, control equipment, and all other significant items.
2. Detailed outline drawings of switchgear showing physical dimensions, weights, and physical location of components and accessories. Also, Manufacturer ’s outline drawing shall provide a material identification number and list of such major components.

3. Nameplate drawing with model or style, electrical ratings and weights in tabular format.
4. One-line, schematics and wiring diagrams for all control, power, and current transformer circuits. Wiring diagrams shall reflect the physical location of components.
5. Current transformer connection tables (for multi-ratio CT's), excitation curves with lead and turn resistance data, and ratio correction curves with ANSI relay or meter accuracy class(es), if applicable.
6. Ratio correction and accuracy curves for all potential transformers.
7. Manufacturer's descriptive data for equipment and devices proposed as substitutions under the "or-equal" clause.
8. AutoCAD™ 2014 .dwg format for all project drawings. Provide files on CD-ROM.

C. Approval Drawing Submittals:

Check shop drawings for errors, correctness of details and conformance with the specifications prior to submittal. Notify Owner of any inconsistencies or questions regarding approval revisions or comments shown on the drawings. Submit all shop drawings to the Owner in the manner specified below.

1. Initial Submittal: Two (2) copies of each shop drawing and any other supplementary data to the Owner at the address given in Section H.2. One copy of the shop drawings indicating the Owner's action will be returned to Manufacturer within two (2) weeks from date of receipt.

The Owner will review all approval drawings for conformity with specifications. Data and drawings will be marked with one of the following responses to indicate result of Owner's review.

2. Reviewed: Indicates that the information or drawing appears to be in general conformity with the specification requirements. Drawings marked "Reviewed" by the Owner may be released for fabrication and/or installation by the Manufacturer.
3. Revised: Owner's comments shall be reviewed and any necessary design changes incorporated into the shop drawings or design before proceeding with fabrication or drawing distribution.
4. Revise and Resubmit: Indicates that the data, drawings, or supplementary information does not appear to be in general conformance with the specifications. Owner's comments will be provided on marked drawings and/or transmitted letter. Manufacturer shall make all necessary revisions and resubmit for Owner's review three (3) copies of all drawings bearing this review note.

Review of shop drawings by Owner does not relieve the Manufacturer of responsibility for errors, correctness of details, or conformance with the specifications.

D. Construction Drawings

Submit to the Owner for distribution six (6) sets of construction drawing prints for switchgear purchased under these specifications. Each set shall be stamped and certified "For Construction". All sets shall be submitted in accordance with date guaranteed by Manufacturer in Proposal Data Sheets.

E. As-Built Documents and Instruction Manuals

Submit to the Owner for distribution, the following as-built documents and bound instruction manuals for switchgear purchased under these specifications:

1. AutoCAD™ 2014 .dwg format for all project drawings. Provide files on CD-ROM.
2. One Electric copy and one printed copy of equipment instruction manuals. Each copy shall thoroughly address equipment installation, operation, and maintenance. Each copy shall also include final test reports, equipment drawings, and renewal parts lists for all replaceable parts and assemblies. Manuals shall be bound by a durable means, properly indexed to identify contents and clearly labeled to indicate Owner, project, and equipment covered.

All copies / sets shall be submitted in accordance with date guaranteed by Manufacturer in Proposal Data Sheets.

1.8 WARRANTY

A. Warranty provided by the Manufacturer shall be understood to cover all defects and problems related to the operation and maintenance of switchgear for a minimum period of one (1) year from the date of commercial operation. Manufacturer guarantees that switchgear is capable of satisfactory and continuous operation in accordance with Manufacturer 's specifications and ratings.

B. Should the Owner detect any defects in design, or any problems with the satisfactory operation of switchgear furnished under these specifications, Owner will notify the Manufacturer of the details related to the defect or problem. The Manufacturer agrees to proceed with diligence in correction of any problems as quickly as possible, particularly where the defect prevents continuous operation of the switchgear. Manufacturer 's guarantee shall cover

the cost of replacement parts and repair, including “in-out” costs of loading, transportation, and off-loading incurred by the Owner.

C. Owner will have right to operate equipment until all defects are corrected, and guarantees met, without cost for depreciation, use or wear.

D. Switchgear may be removed from operation for examination, adjustment, alteration, or change only at time approved by Owner.

1.9 STANDARD PERFORMANCE TESTS

A. Switchgear tests shall be performed in accordance with most current revision of ANSI and IEEE test codes. Manufacturer shall notify Owner two (2) weeks minimum in advance of switchgear testing to allow the standard performance tests to be witnessed by Owner without additional cost to Owner (All transportation and labor costs for Owner’s personnel will be covered by Owner). Test results shall include, but not necessarily be limited to, the following:

1. Hi-potential test on each pole with breaker contacts open.
2. Hi-potential test of switchgear bus assembly.
3. Contact resistance test (micro-ohms) for each pole of each circuit breaker.
4. Timing test for breaker mechanism.
5. Functional and operational tests of all power and control circuits.
6. CT and PT ratio and polarity test to verify integrity of Instrument transformers.

B. All tests shall be performed with installed equipment. Manufacturer shall submit five (5) certified copies of all test data for each unit of switchgear covered by these specifications, to the Owner. Copies shall be included with switchgear instruction manuals or bound in a form suitable for insertion into these manuals by others.

1.10 FIELD SERVING ENGINEER

A. If requested by the Owner as provided for in the Proposal Data Sheets, the Manufacturer shall provide a factory trained Field Service Engineer from the Manufacturer of the switchgear building to supervise inspection, assembly and testing of switchgear at the project site.

B. A visual inspection shall be performed by the Field Service Engineer, including all accessories and parts disassembled and shipped separately to ensure that the equipment was properly handled during transportation. The Field Service

Engineer shall provide all necessary tools and special test equipment necessary to perform standard check out procedures for switchgear and accessories.

C. Field Service Engineer shall also supervise and provide advice on the assembly of switchgear. Field Service Engineer shall have full authority to direct field operations as necessary to assure that switchgear has been assembled and service in accordance with Manufacturer 's requirements. After completion of assembly, check electrical operation of accessories and instruct Owner's personnel on the operation and maintenance of equipment.

D. Cost for field service shall be as provided in the "Proposal Data Sheets". Field service shall be scheduled only at the request of the Owner. If field service trips are necessary for the correction of defects or problems at no fault of the Owner, such trips shall be at Manufacturer 's expense.

1.11 FIELD TESTING BY OWNER

A. The Manufacturer is advised that the Owner may choose to accomplish field tests to confirm any of the shop test results as an independent verification of switchgear's condition. Should the Owner choose to perform such tests, it will be conducted solely at the Owner's expense.

B. Should field tests show discrepancies between Manufacturer 's shop test results and Owner's field tests, or detect any irregularities in the ratings of equipment, Manufacturer shall assist the Owner in interpretation of results and any corrective action necessary to achieve performance guarantees.

1.12 BID EVALUATIONS

A. The Owner will evaluate bids, considering conformance with the specifications, price, delivery, Manufacturer 's reputation, optional services offered (such as field assembly) and other items in the evaluation of proposals. The Owner reserves the right to waive or not waive any irregularities in the bid, to reject any or all bids, and to determine which bid is the lowest and best bid for the installation.

1.13 FAILURE TO MEET GUARANTEED DELIVERY DATE

A. Since delivery of switchgear specified herein is critical to the scheduling of the Owner's project, the Manufacturer shall make every reasonable effort to meet his guaranteed delivery date specified in the Summary of Proposal.

B. The Owner agrees to indemnify the Manufacturer for circumstances beyond the control of the Manufacturer . Circumstances beyond the control of the

Manufacturer are defined as: acts of God, acts of government, transportation to final destination, failure of equipment under test, material shortages due to failure of others to make delivery within the time usually required, and any other circumstances reasonably beyond the control of the Manufacturer .

C. Upon any actual or projected departure from the manufacturing schedule (see Section 1.7A.1) which, in the opinion of the Manufacturer , may result in delay in shipment, the Manufacturer shall immediately notify Owner of the following:
   1. Explanation of delay or potential delay.
   2. Means by which delay could be minimized.
   3. Projected new shipment date.
   4. Additional details as requested by the Owner.

D. Should the Manufacturer fail to make a reasonable effort to meet the guaranteed delivery date where this failure is evidenced by: failure to inform the Owner of changes in the manufacturing schedule, failure to allow sufficient time for delivery of materials and equipment necessary for manufacturing, or lack of cooperation in effective measure by which delays could be minimized, the Owner may penalize the Manufacturer an amount not in excess of $1,000.00 per calendar day for each day that the factory shipping date falls beyond the guaranteed delivery date specified in the Summary of Proposal.

E. The Owner reserves the right to inspect the progress of work at the Manufacturer ’s facilities at any time subsequent to notice of such intent.

1.14 COORDINATION

A. It is the Owner’s intent to use the switchgear building for housing all control and instrumentation equipment for the complete unit substation. The unit substation includes the Switchgear specified herein, external Transformers, and Primary Circuit Breakers. The successful switchgear Manufacturer shall coordinate with the applicable parties to assure proper implementation of all aspects of the complete unit substation. Some of these aspects are (but not limited to): SCADA, relay and CT wiring, overhead bus connection, and AC/DC control and indication requirements.

PART 2 - TECHNICAL REQUIREMENTS

2.1 GENERAL

A. Switchgear covered by these specifications will be installed on Logan City's municipal power system to provide electrical service to residential, commercial and light industrial loads via the City's distribution system.

B. Switchgear will be connected to one substation type transformer by overhead external rigid bus (by Owner). Switchgear shall include exterior roof top bushings and bus support insulators for external buswork to be connected to and supported above the switchgear building.

C. The switchgear described in this specification shall be designed for operation on a three-phase, four-wire, 60-hertz system.

D. The general configuration and certain ratings and requirements for switchgear are depicted on the attached layout drawings, which form a part of this specification.

E. The following technical specification describes minimum requirements and ratings.

2.2 REFERENCE SPECIFICATIONS, CODES AND STANDARDS

A. Industry Codes and Standards

The following Standards and Codes in effect as of the date of this specification, and any subsections thereof as applicable, shall govern design and selection of equipment and material supplied:

1. NEMA (National Electrical Manufacturers Association)
2. IEEE (Institute of Electrical & Electronics Engineers)
    a. IEEE C37.20.2 – Standard for Metal-Clad Switchgear
    b. IEEE C37.100.1 – Standard of Common Requirements for High-Voltage Power Switchgear Rated Above 1000 V
3. ANSI (American National Standards Institute)
    a. ANSI Z535.4 – American National Standard for Product Safety Signs and Labels
4. NEC (National Electric Code)
5. UL (Underwriters' Laboratories)
6. ASTM (American Society for Testing and Materials)

a. ASTM D1535-08 – Standard Practice for Specifying Color by the Munsell System

7. ICC (International Code Council)
   a. IBC – International Building Code

2.3 ENVIRONMENTAL AND SERVICE CONDITIONS

A. Outdoor, continuous duty

B. Approximate altitude above sea level: 4800 Feet

C. Seismic zone: 3

D. Climatic Conditions:

1. Maximum temperature 42° Celsius
2. Maximum average temperature of cooling air for any 24-hour period: 24° Celsius
3. Minimum temperature: -34° Celsius
4. Maximum wind speed: 110 MPH

E. Auxiliary DC power available 48Volts DC

2.4 STATIONARY STRUCTURE

The following shall pertain to all walk-in switchgear covered in these specifications.

A. The general arrangement of the stationary structure shall be as shown on the switchgear plan-view drawing included with these specifications. Alternate configurations shall be proposed at time of bidding, and MAY be considered or rejected at the Owner's discretion. Manufacturer shall include plan view drawings of alternate configurations with bid.

B. The structure shall have a single front aisle designed with the maximum reasonable interior space to permit withdrawal of the circuit breakers for inspection, test, or maintenance, and at a minimum to allow passage of two breakers side by side.

C. Switchgear and aisle portions of the enclosure shall be completely assembled and tested at the factory before shipment. Aisle and switchgear portions shall be ready for easy attachment and final assembly in the field.

D. Approximate location of auxiliary equipment (primary relay panel, DC batteries, battery charger, and RTU) is shown on the layout drawing. Alternate

configurations shall be proposed at time of bidding. Manufacturer shall include plan view drawings of alternate configurations with bid.

E. Switchgear shall have single doors placed at locations shown on the plan-view. Doors shall have the following characteristics:

1. Keyed door handles with ASSA cores approved by Owner. Owner shall provide the keys.
2. Circuit breaker shall be able to pass through the door.
3. Sufficiently braced and reinforced steel to provide a solid contact/seal when closed and avoid warping when opening and closing.
4. Equipped with "full-open" door closers with a latch to hold door open.
5. Panic hardware.

F. Provide hinged, gasketed, full length, weatherproof, rear doors, complete with three-point latch and provisions for padlocking. Hinge shall be on the right side (back view). Each door shall have a sign stating the electrical hazards.

G. All hazard signs shall be in accordance with ANSI Z535.2-2007.

H. The stationary structure for walk-through switchgear shall consist of indoor-class switchgear compartments located in a weatherproof steel housing.

I. The switchgear assembly shall consist of individual vertical sections housing various combinations of circuit breakers and auxiliaries, assembled to form a rigid metal-clad switchgear assembly.

J. All sections housing circuit breakers shall be welded.

K. Metal side sheets shall provide grounded barriers between adjacent structures and solid removable metal barriers shall isolate the major primary sections of each circuit.

L. The structure shall be fabricated from steel sheets and reinforced with structural steel members where necessary:

1. 16" on center interlocking panel construction:
   a. Floor: ¼" steel
   b. Exterior walls: 12 gauge
   c. Interior walls: 16 gauge
   d. Ceiling and roof: as required for design

2. 36" on center welded or bolted panel construction:
   a. Floor: ¼" steel

b. Exterior walls: 11 gauge
c. Interior walls: 14 gauge
d. Ceiling and roof: as required for design

M. The completed switchgear assembly shall be carried on a rigid structural steel framework for bolting to the foundation or piers and to prevent distortion during shipment. Manufacturer shall supply all necessary hardware and clamps for fastening building to foundation if required by design. If design requires embedded anchor bolts only, Owner will supply anchor bolts.

N. Stationary structure shall be insulated to the following minimum values:

1. Walls: R-19
2. Roof: R-30
3. Floor: R-30

O. Foam insulation is preferred.

P. Separate compartments, isolated from each other by sheet steel barriers, shall be provided for the circuit breakers, buses, cable terminals, and instrument transformers. Adequate barriers shall be provided between adjacent housings to prevent interchange of fire or ionized gas between compartments.

Q. Switchgear shall be oriented such that the high voltage power cable exit and the entrance to the switchgear are shown in the preliminary site plan drawing, included with these specifications. Alternate configurations shall be proposed at time of bidding. Manufacturer shall include plan view drawings of alternate configurations with bid.

R. If motor operated mechanisms are used in the circuit breaker compartments for moving the circuit breaker unit from the “disconnected” to the “connected” position and vice-versa, provisions shall be made for manually moving the circuit breaker in the event of a motor failure or the complete loss of power supply to the motors. Mechanical interlocks shall be provided which will positively prevent moving the circuit breaker from one position to another while the breaker is closed.

S. Provide 11-gauge aluminum cable entry plate for feeder cable penetrations to be finalized during installation.

T. Provide adjustable cable supports for incoming cable.

U. Hinged half-length steel panels shall be provided for mounting relays, instruments, meters, and controls. Special attention shall be given to the adequate isolation of low voltage control circuits from the high voltage circuits.

V. The horizontal centerline from the top row of relays shall not be placed more than 72 inches off the floor.

W. Control switches shall not be placed more than 72 inches off of the floor.

X. When available as an option, all panel-mounted devices shall be draw out type.

Y. All breakers shall be designed to roll directly from cubicle to switchgear floor without requiring lifts, ramps or jacks.

Z. All compartments shall have enough space to allow racking the breakers to the "disconnect" position with the door closed.

AA. Provisions for breaker remote racking systems shall be included in the bid, but priced separately as optional equipment. It is likely that remote racking will be included so please include this in your bid.

BB. All steel shall be thoroughly chemically cleaned and treated. After cleaning, all metal surfaces shall be inspected for rust and scale, and if necessary, further cleaned to remove all rust and scale. The following paint details are required:

1. Exterior color shall be ANSI #70 light gray, 5 mil minimum.
2. Structure interior walls: white.
3. Structure interior floor: gray with long-wearing non-skid surface.
4. Switchgear:
    a. Fronts: ANSI #70
    b. Interior of cubicles: white (including any and all back panels/pans used for mounting equipment and accessories
    c. Base: corrosion resistant undercoating.

2.5 AUXILIARY EQUIPMENT

A. External control cables for lighting and power circuits, control circuits, current transformer circuits, and alarm circuits will enter from below. Provide a field termination cabinet within the building with all necessary terminal blocks and a minimum of 20% spare 12 pole terminal blocks for field cable connections.

B. Provide adequate lighting in the interior, with switches located near all access doors.

C. Provide exterior rated photo eye LED outdoor lights over each door.

D. Provide a minimum of six (6) utility duplex receptacles equally spaced for convenient use inside the building. Outside the building, provide two (2) 120 V receptacles near each access door.

E. All current transformers, including auxiliary CT's shall be connected to shorting type terminal blocks GE-Type EB27 or equal. Interface with customer's external wiring for control and CT circuits shall be made on terminal blocks. At least 10 percent spare terminals on terminal blocks shall be provided for customer's future use. All CT leads shall be terminated for multi-ratio CT's.

F. Each main bus shall have three (3) voltage transformers, 7,200 – 120 volt with trunnion-mounted high interrupting capacity current limiting fuses wye connected to the 12.47 kV bus for metering. ANSI accuracy class 0.3 for W-Z burdens.

G. Each Feeder Breaker Cubicle: One (1) set – Current transformer, 1200/5 ampere multi-ratio, single secondary, 15 kV, ANSI relaying accuracy class, C200 for 12.47 kV feeder relay located on the bus side of the circuit breaker.

H. Each Main Breaker Cubicle: One (1) set – Current transformer, 2000/5 ampere fixed-ratio, single secondary, 15 kV, ANSI relaying accuracy class, C200 for transformer ddifferential relay located on the bus side of the circuit breaker

I. Each Breakered Cubicle: One (1) set – Current transformer, 2000/5 ampere fixed-ratio, single secondary, 15 kV, ANSI relaying accuracy class, C200 for 12.47 kV bus differential relay located on the load or transformer side of the circuit breaker

J.

K. Provide thermostatically controlled strip heater with perforated shields in each circuit breaker cubicle for anti-condensation. Size as required ambient temperature ranges specified.

L. Provide a switchable LED lamp and duplex outlet in all control cubicles. Locate light switch near receptacle.

M. AC and DC voltage for the switchgear, transformer, and gas circuit breaker will be supplied from the switchgear structure, at 120/240 volts AC and 48 volts DC.

N. AC Distribution Panelboard shall be one (1) indoor NEMA 12 AC panelboard, Square D-type NQOD, 42 circuit, 120/240 V, 60 hertz, single phase-three wire, 225A, thermal magnetic main breaker, and Square D Type QO branch breakers.

1. Main and branch breakers shall be rated 10 kAIC minimum. Enclosure shall be surface mounted at location shown on 15 kV Metal-clad Switchgear Arrangement Drawing.
2. Panelboard shall be labeled “AC LOADCENTER”; nameplate shall be black on white with 3/8” lettering.

O. DC Distribution Panelboard shall be one (1) indoor NEMA 12 panelboard, bottom feed with 225A main, 100A Thermal Magnetic main Breaker and branch breakers. **Panel and Breakers MUST be fully rated at 48 VDC**:

1. Five (5) -20A double pole
2. Ten (10) -30A double pole
3. Five (5) -40A double pole
4. Main and branch breakers shall be rated 5 kAIC minimum. Enclosure shall be surface mounted at location shown on 15kV Metalclad Switchgear Plan View Drawing.
5. Panelboard shall be labeled “DC LOADCENTER”; nameplate shall be white on black with 3/8” lettering.

P. Manufacturer shall pre-wire all AC lights, heaters, and circuit breaker power and control circuits. Wiring to be routed in permanent conduit and tray separating low voltage SCADA circuits from high voltage circuits (120 V and above) for a neat installation.

Q. Provide wireway and conduit as required for Owner’s installation of external cables (spare AC and DC circuit breakers). Wireway shall be installed along interior walls, near the ceiling to facilitate field wiring of additional equipment by Owner.

R. Manufacturer shall prewire for communication between IEDs, CTs, etc. and Protective Relays.

Manufacturer

S. HVAC

1. Manufacturer shall provide one (1) exterior wall mounted self-contained heating and air-conditioning units, complete with supply and return grills, barometric fresh air damper and disposable air flitter to be controlled by a programmable thermostat. BARD units are preferred.
2. HVAC unit shall be independently wired through externally mounted lockable disconnect switches.
3. HVAC unit shall be on a dedicated circuit from the AC panel.

4. The Manufacturer shall size this unit based on the heat gain of the equipment installed, the environmental conditions and the required operating range.
5. The HVAC unit shall be positioned to provide adequate air distribution throughout the switch gear assembly.

2.6 ELECTRICAL BUSES AND SUPPORTS

A. Shall be of high conductivity silver-plated copper with adequate current carrying capacity.

B. Shall have fluidized bed or powder coat epoxy flame-retardant and track-resistant insulation. (Sleeving not allowed).

C. Circuit breaker primary spouts shall be porcelain.

D. Bus shall have provisions for future extension.

E. The temperature rise of the bus and connections shall be in accordance with ANSI standards and documented by design tests.

F. Bus joints and tap joints shall be silver plated and tightly bolted with Belleville washers to insure a good connection. Joints shall be insulated with easily installed boots. Provision shall be made and materials supplied to complete the insulation for all joints and taps which must be made in the field to complete the assembly.

G. The main horizontal bus and all vertical buses shall be of non-tapered construction braced to withstand a short circuit rating of 40 kA RMS symmetrical. All buses and taps shall be completely insulated and rated 110 kV BIL. Other bus ratings include:

1. Main bus: 2000A continuous.
2. Individual feeder buses: 1200A continuous.

H. Ground bus:

1. Switchgear structure and base shall be bonded together for grounding. Provide NEMA 2-hold ground pads on exterior base corners of switchgear for tie to substation ground grid.
2. Equipment grounds: Provide dedicated, pre-drilled, perimeter copper ground bus wall mounted above wireway and switchgear control panels for AC panel, DC panel, battery charger, RTU, and relay panels. Bond all dedicated ground bus with 2/0 soft drawn, green insulated copper. If

ground bus is not within a cabinet, place no more than four (4) feet from equipment.

3. A continuous silver plated copper ground bus measuring ¼" X 2" shall be provided for the full length of the switchgear assembly. The ground bus shall be capable of carrying the rated short circuit current of the installed circuit breakers for a minimum of 2 seconds.

I. Power bus orientation shall be A-B-C top to bottom, front to back, and left to right when viewed from the front of the switchgear.

2.7 VACUUM CIRCUIT BREAKERS

A. The circuit breaker units shall be electrically operated, three-pole, draw out type, rated 15 kV (minimum), 110 kV BIL, 60 Hz, with primary and secondary disconnecting devices. Interlocks shall be provided where specified.

B. A device shall automatically open the primary disconnecting devices when the circuit breaker reaches the "disconnect" position.

C. Primary and secondary disconnecting devices shall have silver contacts designed for cool operation with minimum maintenance. Guides shall be provided to insure proper alignment of primary and secondary disconnecting devices before engagement.

D. All unused "a" and 'b" auxiliary switch contacts available on each breaker shall be wire to a terminal block where they will be readily available for use by the Owner. Manufacturer is to provide a minimum of six (6) unused "a" and six (6) unused "b" auxiliary switch contacts for each breaker.

E. Auxiliary Truck Operated Cell switches (TOC) shall be provided to indicate circuit breaker element position.

F. Provisions for padlocking circuit breakers in the disconnected position shall be provided to accommodate standard lockout tagout procedures.

G. Each circuit breaker shall have the following minimum ratings:

1. Normal voltage: 12.47 kV
2. Rated Maximum Voltage (rms): 15 kV
3. Rated Voltage Range Factor (K): 1.30
4. Insulation level (Nominal Freq.): 36 kV rms
5. Insulation Level (1.2 X 50 µsec): 110 kV crest
6. Rated Continuous Current:
   a. Mains: 2000A

b. Feeders: 1200A

7. Rated Short Circuit Current: 18 kA rms
8. Maximum Symmetrical Interrupting Capability: 23 kA rms
9. Closing and latching Capability: 37 kA rms (asymmetrical)
10. Clearing Time: 5 cycles

H. The circuit breaker units shall be mechanically trip free, stored-energy type, with 48 volt DC trip and close mechanism.

I. The breaker shall be complete with necessary auxiliary switches, latch checking switch, control relay and operation counter, and mechanical interlocks to prevent removal of breaker unless it is in open position.

J. The breaker shall have a minimum of four mechanically operated switch contacts (MOC) (two (2) normally open when breaker is open, two (2) normally closed when breaker is open) wired to terminal blocks for customer connections.

K. The breaker stationary unit shall have a set of primary disconnecting devices with automatic shutters and a set of secondary disconnecting devices.

2.8 ELECTRICAL CONTROL SPECIFICATIONS

A. Panelboard wire shall be extra flexible, 41 strand tinned copper conductor, rated 600 volt, Type SIS #14 AWG, U.L. listed, except CT circuits where #10 AWG Type SIS, 65 strand shall be used. All indication wiring for SCADA shall use #18 AWG.

B. Wire terminals shall be insulated barrel, color coded, tin plated copper, ring type with appropriate ring size for mating stud, and barrel size, for conductor.

C. All device stud connections shall be made with lock washers.

D. Any one (1) terminal board point shall be terminated with no more than one (1) control cable conductor. Any one (1) screw connection shall be terminated with no more than two (2) wires and any one (1) stud or bolted connection terminated with no more than three (3) wires.

E. Each terminal block, conductor, device, fuse block, and terminal shall be permanently labeled to coincide with the identification shown on the drawings.

F. All conductors shall be identified using heat shrink destination wiring labels. Terminal and conductor identification shall be by a permanent method

unaffected by heat, solvents, or steam and not easily dislodged. Adhesive labels are not acceptable.

G. All wiring shall be installed in a workmanship-like manner. As required to provide a neat appearance, wires shall be bundled by use of nylon cable ties.

H. Support in routing of conductors via wire duct shall be Panduit, Panduct, or equal. Screw applied tie mounts are an acceptable alternative. Self-adhesive or epoxy tie mounts are unacceptable.

I. All connections between devices on a switchboard and any devices external to the switchboard shall be made via terminal boards. All termination shall be made with ratcheted compression tools so as to assure physically and electrically secure terminations.

J. Terminal boards shall be rated 20A, 150° C, 600 volt and shall be one (1) piece molded phenolic with barriers and strap screw connections. Twelve (12) point terminal boards shall be GE type EB-25, or equivalent, with white marking strips. Shorting strip terminal boards used for termination of current circuits shall be six point GE type EB-27, or equivalent.

K. Spare terminals shall be provided on each panelboard to the extent of 10% used or a minimum of one spare twelve (12) point terminal board, whichever is greater.

L. A field termination cabinet shall be provided as indicated on the drawings. All control, power, and indication wiring to and from field equipment shall be terminated in the field termination cabinet. All field CT leads shall terminate on CT shorting blocks. A minimum 20% spare terminals for both twelve (12) point terminal boards and CT terminal boards.

M. Fuse holders shall be 250 volt, rating and number of poles as specified and shall be of porcelain construction with side barriers.

N. One (1) time fuses shall be 250 volt, Class H, 10,000A RMS symmetrical interrupting rating and shall be UL listed. Supply Bussman Type NON, or equal.

O. Nameplates shall be provided for each device as listed on the nameplate list (to be provided by Engineer at a later date). Nameplates shall be of white plastic construction with black lettering. Size and lettering of engraving shall be as specified. Nameplates shall be attached to panels with stainless steel screws.

P. All devices including those not requiring nameplates, such as terminal boards and fuses, shall be labeled on the rear of the panel. Labels shall be self-

adhesive white markers, type written with device designations corresponding to those used on wiring diagrams.

Q. Nominal control voltage will be 48 volts DC

R. Protective relays, draw-out case devices, recording instruments, meter, test switches, and all front panel mounted devices shall be panel mounted and rear connected unless otherwise specified. Extended panel mounts are acceptable if necessary to accommodate breaker racking operations.

S. Input and output surge ratings of protective relays and communications equipment shall be in accordance with ANSI/EEEE Standard C37.90.1 – (latest revision). Dedicated case grounds shall be installed.

T. Control and instrument switches shall be rated 600 volt, 20A, Electroswitch Series 24 with contacts and handle as specified, or equal.

U. Indicating lamps shall be LED and rated 48 VDC.

2.9 SWITCHGEAR LINEUP DETAILS

A. In the event that certain items are indicated on the drawings but not tabulated on the itemized list, or vice versa, Manufacturer shall use the total quantity of equipment to satisfy the requirements of both the drawings and the list. All miscellaneous items and accessories required in conjunction with any of the items listed or shown on the drawings shall be furnished by the Manufacturer .

B. The line-up shall of switchgear assembled to be field connected through underground cables as described in this specification. Each unit shall have a 120 V receptacle with a switched light.

C. The following items shall be pre-wired and/or provisions made to facilitate the implementation of SCADA. All control wiring shall be #14 AWG with indication wiring #18 AWG.

D. Accessories and Spare Parts

1. Provisions for breaker remote racking systems shall be included in the bid, but priced separately as optional equipment.
2. Two (2) – Spare fuses of each and all types used
3. One (1) – Breaker transfer truck and inspection rack

4. One (1) – Breaker test panel, secondary disconnect cable, and on/off switch to electrically test breaker operation while performing breaker maintenance.

2.10 RELAYS AND SWITCHES

Description	Relay	Part Number	Quantity
12kV Bus Differential	SEL-587Z	0587Z0X225312XX	One (1)
Breaker Overcurrent	SEL-351S	0351S5X3D3E52X1	Five (5)
Serial Port Switch	SEL-3610	3610X3B1XXX0	One (1)
Cutout Switch	ElectroSwitch	24203B	One (1)
Lockout Relay	ElectroSwitch	7808C	One (1)
LED Indicating Light (B)	General Electric	0116B6708G2-B	Two (1)
Test Switch	FT1	ABB P,P,P,C-C,C-C,C-C,P C129A514G01	Six (6)
Test Switch	FT1	ABB T,T,T,T,T,T,T,T,T,T C129A539G01	Six (6)
Temperature Monitor	ARGA	ARGA 26-963-48V-J3.0DC Voltage/Temperature Transducer	One (1)
Comm. Cable	SEL-273A	Length as needed	Six(6)

PART 3 - PROPOSAL

3.1 PROPOSAL DATA SHEETS

A. Total firm price to furnish and deliver F.O.B., jobsite, Logan, Utah, one (1) 15' X 28' walk-through, 15kV outdoor metal-clad overhead bus connected switchgear line-up as described by these specifications. $ _____

B. Alternate Bid 1: Building Dimensions _____ X _____ (Attach proposed layout drawing) $ _____

C. Shipping and Offloading $ _____

D. Remote Racking System:

1. Provisions and equipment required to rack circuit breakers from a remote location. $ _____

E. Delivery Date:

1. *Manufacturer* shall enter the earliest guaranteed delivery date of equipment to its final destination. _____ (Month)/(Day)/(Year)

F. Cancellation Charges:

Bidders quoted cancelation charge at 90 days will be considered in the bid evaluation. (Maximum cancellation charges from date of award % of bid price)

1.	30 Days	_____ %
2.	60 Days	_____ %
3.	90 Days	_____ %
4.	120 Days	_____ %
5.	150 Days	_____ %
6.	180 Days	_____ %

3.2 SERVICE ENGINEER (OPTIONAL)

The *Manufacturer* proposes the following to supervise equipment field assembly and installation:

A. Daily Rate (including all expenses): $ _____

B. Rate for one round trip to the jobsite, including all expenses: $ _____

C. Total estimated number of working days required for Service Engineer _____ days

D. Do prices quoted in include Field Service Engineer? _____ (yes, no)

1. If so, how many days: _____ days

3.3 EXTENDED WARRANTY (OPTIONAL)

A. Three Year (from delivery) $ _____

B. Five Year (from delivery) $ _____

3.4 DRAWINGS

A. Approval Drawings

1. Outline Drawings _____

a. Guaranteed Date _____ (Month)/(Day)/(Year)

2. All drawings except outline drawings

a. Guaranteed Date _____ (Month)/(Day)/(Year)

B. Certified Construction Drawings

a. If no, Guaranteed Date _____ (Month)/(Day)/(Year)

C. Instruction Manuals

a. Guaranteed Date _____ (Month)/(Day)/(Year)

3.5 DATA

A. Manufacturer _____

B. Location of Factory _____

C. Physical dimensions of each circuit breaker cubicle _____ (Width x Depth) (Inches)

D. Estimated width x length of longest piece for shipment _____ (Width x Length) (Inches)

E. Overall switchgear dimensions _____ (Width x Length x Height) (Inches)

F. Proposed CT relaying accuracy class.

1. 2000:5 single-ratio _____
2. 1200:5 multi-ratio _____

G. Bus Material _____

H. Main Bus Cross Section Dimensions _____ (Inches x Inches)

I. Ground Bus Cross Section Dimensions _____ (Inches x Inches)

J. RMS Symmetrical Rating (ie. Bus Bracing) _____ Amps

K. Maximum temperature rise of a 40°C ambient _____ °C

3.6 2000 AMP MAIN BREAKER DATA

A. Manufacturer and Model/Type _____

B. Maximum Rated Voltage _____ Volts

C. Basic Impulse Insulation Level (BIL) _____ kV

D. Maximum temperature rise over a 40°C Ambient _____ °C

E. Continuous Current Rating _____ Amps

F. Short Time Symmetrical Rating _____ Amps

G. Recommended Maintenance Interval (Number of Rated Fault Interruptions) _____

H. Rated Breaker Interrupting Time _____ Cycles

I. Charging Spring Motor

1. Voltage Rating _____ Volts
2. Inrush _____ Amps
3. Running _____ Amps

J. Peak Tripping Coil Current _____ Amps

K. Peak Closing Coil Current _____ Amps

3.7 1200 AMP MAIN BREAKER DATA

A. Manufacturer and Model/Type _____

B. Maximum Rated Voltage _____ Volts

C. Basic Impulse Insulation Level (BIL) _____ kV

D. Maximum temperature rise over a 40°C Ambient _____ °C

E. Continuous Current Rating _____ Amps

F. Short Time Symmetrical Rating _____ Amps

G. Recommended Maintenance Interval (Number of Rated Fault Interruptions) _____

H. Rated Breaker Interrupting Time _____ Cycles

I. Charging Spring Motor

1. Voltage Rating _____ Volts
2. Inrush _____ Amps
3. Running _____ Amps

J. Peak Tripping Coil Current _____ Amps

K. Peak Closing Coil Current _____ Amps

3.8 ASSEMBLY

Please provide a detailed description of the assembly required for the 15kV walk-through switchgear:

_____

_____

_____

_____

_____

3.9 EXCEPTIONS

Please list any exceptions to the requirements in this specification:

_____

_____

_____

_____

_____

3.10 AUTHORIZED SIGNATURE

A. Seller: _____

B. Signature & Title: _____

C. Address: _____

_____

D. Date:_____

PART 4 - CITY OF LOGAN INSURANCE AND BOND REQUIREMENTS

FOR: Substation 6 Arc-Resistant 15kV Switchgear

The Contracting party shall procure and maintain for the duration of the contract insurance and bonds against claims or liability which arises out of or in connection with the performance of the work hereunder by the Contracting party, his agents, representatives, employees or subcontractors. The cost of such insurance and bonds shall be included in the Contracting party's bid or proposal.

A. TYPES AND MINIMUM LIMITS OF INSURANCE

Contracting party shall maintain limits not less than:

1. **GENERAL LIABILITY**: $4,000,000 combined single limit per occurrence, personal injury and property damage, $5,000,000 aggregate. Broad Form Commercial General Liability is required. (ISO 1993 or better) to include Products - Comp/OP aggregate of $4,000,000. Limits to apply to this project individually.

2. **PROFESSIONAL LIABILITY:** Not applicable.

3. **AUTOMOBILE LIABILITY**: $2,000,000 per occurrence. "Any Auto" coverage is required.

4. **WORKERS' COMPENSATION and EMPLOYERS LIABILITY**: Workers' Compensation statutory limits as required by the Workers Compensation Act of the State of Utah and Employers Liability limits at a minimum of $100,000 per occurrence.

5. **PAYMENT and PERFORMANCE BONDS:** Contracting party shall provide payment and performance bonds in a form acceptable to the City and in the full amount of the contract.

B. DEDUCTIBLES AND SELF-INSURED RETENTIONS

Any deductibles or self-insured retention exceeding 5% of the policy limits must be declared to and approved by Logan City. At the option of Logan City, either (1) the insurer may be required to reduce or eliminate such deductibles or self-insured retention as respects Logan City, its officers, officials and employees; or (2) the Contracting party may be required to procure a bond guaranteeing payment of losses and related investigations, claim distribution and defense expenses.

C. NOTICE OF ACCIDENT, INJURY, OR DAMAGE

Contracting party shall agree to promptly disclose to Logan City, all incidents or occurrences of accident, injury, and/or property damage covered by the insurance policy or policies.

D. OTHER INSURANCE PROVISIONS

The policies are to contain, or be endorsed to contain, the following provisions:

I. General Liability and Automobile Liability Coverages

A. Logan City, its officers, officials, employees and volunteers are to be covered as additional insureds as respects: liability arising out of activities performed by or on behalf of the contracting party; products and completed operations of the Contracting party; premises owned, leased, hired or borrowed by the Contracting party. The coverage shall contain no special limitations on the scope of protection afforded to Logan City, its officers, officials, employees or volunteers.

B. The Contracting party's insurance coverage shall be a primary insurance as respects to Logan City, its officers, officials, employees and volunteers. Any insurance or self-insurance maintained by Logan City, its officers, officials, employees or volunteers shall be in excess of the Contracting party's insurance and shall not contribute with it.

C. Any failure to comply with reporting provisions of the policies shall not affect coverage provided to Logan City, its officers, officials, employees or volunteers.

D. The Contracting party's insurance shall apply separately to each insured against whom claim is made or suit is brought, except with respects to the limits of the insurer's liability.

E. ACCEPTABILITY OF INSURERS

Insurance and bonds are to be placed with insurers admitted in the State of Utah with an A. M. Best rating of not less than A-: IX, and in the limits as listed in this document, unless approved by the City's Risk Manager, or his designee, **a minimum of five (5) business days prior to bid or proposal deadline.**

F. VERIFICATION OF COVERAGE

Contracting party shall furnish Logan City with certificates of insurance and with original endorsements effecting coverage required by this clause. The certificates and endorsements are to be signed by a person authorized by that insurer to bind coverage on its behalf. The certificates and endorsements are to be on forms acceptable to Logan City before work commences. Logan City reserves the right to require complete, certified copies of all required insurance policies, with all endorsements, at any time.

G. SUBCONTRACTORS

Contracting party shall include all subcontractors as insureds under its policies or shall furnish separate certificates and endorsements for each subcontractor. All coverages for subcontractors shall be subject to all of the requirements stated herein.

PART 5 - CONTRACT FOR CONSTRUCTION

This Contract is by and between The City of Logan (hereinafter called the OWNER) and _____ _____ (hereinafter called MANUFACTURER). OWNER and MANUFACTURER, in consideration of the mutual covenants hereinafter set forth, agree as follows:

ARTICLE 1 – THE PROJECT

1.01 The Project for which the Work under the Contract Documents shall apply is generally described as follows:

Manufacture, supply, and install Arc-Resistant 15kV Switchgear in Substation 6

ARTICLE 2 – WORK

2.01 MANUFACTURER shall complete all Work as specified or indicated in the Contract Documents for completion of the project.

ARTICLE 3 – ENGINEER

3.01 The ENGINEER, unless otherwise indicated in the Contract Documents, shall be the CITY ENGINEER, or his appointed representative.

ARTICLE 4 – CONTRACT TIMES

4.01 Time of the Essence:

A. All time limits for Milestones, if any, Substantial Completion, and completion and readiness for final payment as stated in the Contract Documents are of the essence of the Contract.

4.02 Dates for Substantial Completion and Final Payment:

A. The Work will be substantially completed on or before _____ and completed and ready for final payment in accordance with the General Conditions.

4.03 Liquidated Damages:

A. MANUFACTURER and OWNER recognize that time is of the essence of this Agreement and the OWNER will suffer financial loss if the Work is not completed within the times specified in paragraph 4.02, plus any extensions thereof allowed in accordance with the General Conditions. The parties also recognize that it will be impracticable to determine actual damages which OWNER will sustain in the event of or by reason of the delay. Accordingly, instead of requiring any such proof, OWNER and MANUFACTURER agree that as liquidated damages for delay (but not

as a penalty) MANUFACTURER shall pay OWNER $1,000.00 for each calendar day that expires after the specified time in paragraph 4.02 for substantial completion until the Work is substantially complete. After substantial completion, if MANUFACTURER shall neglect, refuse, or fail to complete the remaining Work within the contract time or any proper extension thereof granted by OWNER, MANUFACTURER shall pay OWNER $1,000.00 for each calendar day that expires after the time specified in paragraph 4.02 for completion and readiness for the final payment until the Work is completed and ready for final payment. It is further agreed that the amount stipulated for liquidated damages per day of delay is a reasonable estimate of the damages that would be sustained by OWNER, and MANUFACTURER agrees to pay such liquidated damages as herein provided. In case the liquidated damages are not paid, MANUFACTURER agrees that OWNER may deduct the amount thereof from any money due or that may become due to MANUFACTURER by progress payments or otherwise under the Agreement, or if said amount is not sufficient, recover the total amount.

ARTICLE 5 – CONTRACT PRICE

5.01 OWNER shall pay MANUFACTURER FOR COMPLETION OF THE Work in accordance with the Contract Documents an amount in current funds equal to the sum of the amounts determined pursuant to paragraph 5.01.A below:

A. For all Work, the sum of:

_____ $_____

(use words) (use figures)

ARTICLE 6 – PAYMENT PROCEDURES

6.01 Submittal and Processing of Payments:

A. MANUFACTURER shall submit Applications of Payment in accordance with the General Conditions. Applications for Payment will be processed by ENGINEER as provided in the General Conditions.

6.02 Progress Payments; Retainage:

A. OWNER shall make progress payments on account of the Contract Price on the basis of MANUFACTURER ’s Applications for Payment once each month during performance of the Work as provided in paragraphs 6.02.A.1 and 6.02.A.2 below. All such payments will be measured by the Schedule of Values established in the General Conditions (and in the case of Unit Price Work based on the number of units completed) or in the event there is no Schedule of Values, as provided in the General Requirements:

1. Prior to Substantial Completion, progress payments will be made in an amount equal to the percentage indicated below but, in each case, less the aggregate of

payments previously made and less such amounts as ENGINEER may determine or OWNER may withhold, in accordance with the General Conditions:

a. The OWNER will pay the MANUFACTURER a progress payment on the basis of the approved partial payment estimate submitted by the MANUFACTURER and verified by the ENGINEER, including any approved Change Orders, but will not hold retainage.

2. Upon Substantial Completion of the project, the OWNER will pay to 95 percent of the revised contract amount including any approved Change Orders.

6.03 Final Payment

A. When all items on any generated 'Punch' or Completion List have been completed, the OWNER will pay the remaining 5 percent of the revised contract amount within 30 days of notification to the OWNER in writing by the MANUFACTURER that all items have been completed. The OWNER shall verify that all items have been completed before final payment is made.

ARTICLE 7 – INTEREST

7.01 All moneys not paid when due as provided in the General Conditions shall bear interest at the same rate that OWNER's funds accrue interest in the State's treasury account.

ARTICLE 8 – MANUFACTURER 'S REPRESENTATIONS

8.01 In order to induce OWNER to enter into this Agreement, MANUFACTURER makes the following representations:

A. MANUFACTURER has examined and carefully studied the Contract Documents and the other related data identified in the Bidding Documents.

B. MANUFACTURER has visited the Site and become familiar with and is satisfied as to the general, local, and Site conditions that may affect the cost, progress, and performance of the Work.

C. MANUFACTURER is familiar with and is satisfied as to all federal, state, and local Laws and Regulations that may affect cost, progress, and performance of the Work.

D. MANUFACTURER has carefully studied all: (1) reports of explorations and tests of subsurface conditions at or contiguous to the Site and all drawings of physical conditions in or relating to existing surface or subsurface structures at or contiguous to the Site (except Underground Facilities) which have been identified in the Supplementary Conditions as provided in the General Conditions and (2) reports and drawings of a Hazardous Environmental Condition, if any, at the Site which is identified in the Supplementary Conditions as provided in the General Conditions..

E. MANUFACTURER has obtained and carefully studied (or assumes responsibility for having done so) all additional or supplementary examinations, investigations, explorations, tests, studies, and data concerning conditions (surface, subsurface, and Underground Facilities) at or contiguous to the Site which may affect cost, progress, or performance of the Work or which relate to any aspect of the means, methods, techniques, sequences, and procedures of construction to be employed by MANUFACTURER , including applying the specific means methods, techniques, sequences, and procedures of construction, if any, expressly required by the Contract Documents to be employed by MANUFACTURER , and safety precautions and programs incident thereto.

F. MANUFACTURER does not consider that any further examinations, investigations, explorations, tests, studies or data are necessary for the performance of the Work at the Contract Price, within the Contract Times, and in accordance with the other terms and conditions of the Contract Documents.

G. MANUFACTURER is aware of the general nature of work to be performed by OWNER and others at the Site that relates to the Work as indicated in the Contract Documents.

H. MANUFACTURER has correlated the information known to MANUFACTURER, information and observations obtained from visits to the Site, reports and drawings identified in the Contract Documents, and all additional examinations, investigations, explorations, tests, studies, and data with the Contract Documents.

I. MANUFACTURER has given ENGINEER written notice of all conflicts, errors, ambiguities, or discrepancies that MANUFACTURER has discovered in the Contract Documents, and the written resolution thereof by ENGINEER is acceptable to MANUFACTURER.

J. MANUFACTURER agrees to register and participate in the Status Verification System to verify the work eligibility status of the MANUFACTURER ’s new employees that are employed in the state as set forth in Utah Code Section 63G-12-302. Each Manufacturer or sub Manufacturer who works under or for another Manufacturer shall certify to the main Manufacturer by affidavit that the Manufacturer or sub Manufacturer has verified through the Status Verification system the employment status of each new employee of the respective Manufacturer or sub Manufacturer.

K. The Contract Documents are generally sufficient to indicate and convey understanding of all terms and conditions of performance and furnishing of the work.

ARTICLE 9 – CONTRACT DOCUMENTS

9.01 Contents:

A. The Contract Documents consist of the following:
   1. Instructions to Bidders
   2. Bid Bond
   3. This Agreement
   4. Performance Bond
   5. Payment Bond
   6. General Conditions, Document 007200 of the APWA Manual of Standard Specifications, 2007 Edition, as modified herein.
   7. Special Provisions
   8. APWA Manual of Standard Specifications, including Standard Plans
   9. City of Logan Standards and Specifications, including Standard Plans
   10. Construction Plans
   11. Addenda (numbers _____ to _____ inclusive)
   12. Insurance and Bond Requirements
   13. Exhibits to this Agreement (enumerated as follows):
       a. Proposal
       b. Bid Schedule
       c. Schedule of Values (for Lump Sum Bid)
       d. Measurement and Payment
       e. Schedule of Project Completion
       f. Manufacturer Qualification Form
       g. Certificate of Insurance, approved by OWNER's Risk Management Division
   14. The following which may be delivered or issued on or after the Effective Date of the Agreement and are not attached hereto:
       a. Notice of Award
       b. Notice to Proceed
       c. Written Amendments
       d. Change Orders
       e. Field Orders

B. There are no Contract Documents other than those listed above in this Article 9.

C. The Contract Documents may only be amended, modified or supplemented as provided in the General Conditions.

ARTICLE 10 – MISCELLANEOUS

10.01 Terms:

Terms used in this Agreement will have the meanings indicated in the General Conditions

10.02 Assignment of Contract:

No assignment by a party hereto of any rights under or interests in the Contract will be binding on another party hereto without the consent of the party sought to be bound; and specifically but without limitation, moneys that may become due and moneys that are due may not be assigned without such consent (except to the extent that the effect of this restriction may be limited by law), and unless specifically stated to the contrary in any written consent to an assignment, no assignment will release or discharge the assignor from any duty or responsibility under the Contract Documents.

10.03 Successors and Assigns:

OWNER and MANUFACTURER each binds itself, its partners, successors, assigns, and legal representatives to the other party thereto, its partners, successors assign and legal representatives in respect to all covenants, agreements, and obligations contained in the Contract Documents.

10.04 Termination for OWNER's Convenience:

A. Upon ten (10) days' written notice to Manufacturer, Owner may, without cause and without prejudice to any other right or remedy of Owner, terminate the Contract. In such case, Manufacturer shall be paid for (without duplication of any payment amounts or work items):

1. completed and acceptable Work executed prior to the effective date of termination, in accordance with the payment provisions of the Contract Documents; provided, however, that no completed work items shall be paid on a cost-plus basis;

2. reasonable expenses sustained prior to the effective date of termination in performing services and furnishing labor, materials, or equipment as required by the Contract Documents in connection with uncompleted Work or unacceptable Work (but only those portions of the unacceptable Work that are acceptable and Manufacturer shall not be entitled to any compensation for any portions of the Work which are unacceptable), plus fair and reasonable sums for overhead and profit on such expenses; provided, however, that Manufacturer shall not be entitled to additional compensation for Work items paid under subparagraph 10.04 A.1. above;

3. costs incurred in settlement of terminated contracts with Sub Manufacturer s, Suppliers, and others; provided, however, that Manufacturer shall not be entitled to additional compensation for Work items paid under subparagraphs 10.04 A.1. and 2. above; and

4. reasonable expenses directly attributable to termination; provided, however,

that Manufacturer shall not be entitled to additional compensation for Work items paid under subparagraphs 10.04 A. 1., 2. and 3. above.

B. Manufacturer shall not be paid on account of loss of anticipated profits or revenue or other economic loss arising out of or resulting from such termination. Termination under this Subparagraph 10.04 shall not entitle Manufacturer to compensation on a cost-plus basis except for items paid under Subparagraph 10.04 A. 2. above.

C. This Article 10.04 shall amend the General Conditions, Document 007200 referenced in Article 9.01 A. 6. by replacing the "Termination for Owner's Convenience" clause contain in 15.2 F. therein.

10.05 Severability:

Any provision or part of the Contract Documents held to be void or unenforceable under any Law or Regulation shall be deemed stricken, and all remaining provisions shall continue to be valid and binding upon OWNER and MANUFACTURER, who agree that the Contract Documents shall be reformed to replace such stricken provision of part thereof with a valid and enforceable provision that comes as close as possible to expressing the intention of the stricken provision.

10.06 Guarantee Period

The MANUFACTURER agrees to indemnify and save harmless the OWNER from any and all defects appearing or developing in the workmanship or materials performed or furnished under the Contract for a period of **one (1) year** after the date of the written notice from the Engineer recommending final acceptance of the entire project by the OWNER.

10.07 Counterparts

This agreement may be executed in counterparts, each of which shall be deemed an original but all of which shall constitute one and the same instrument.

IN WITNESS WHEREOF, the Parties have duly caused this Agreement to be executed on their respective behalves. This Agreement will be effective on _____.

OWNER: MANUFACTURER:

City of Logan	
By:	By:
Attest:	Attest:
Address for giving notices:	Address for giving notices:
290 North 100 West	
Logan UT 84321	

END OF DOCUMENT

PART 6 - ATTACHMENTS

Purchase Order Terms and Conditions of the City of Logan "City"

All special conditions on the face of the purchase order are hereby made a part of the general terms and conditions.

1. COMMERCIAL DETAILS.
Invoice Address. To ensure prompt Payment mail invoices to:

City of Logan
290 North 100 West
Logan, Utah 84321
Attention: Accounts Payable

Tax exemptions. By statute the City is exempt from state and local taxes.

Goods Rejected. GOODS REJECTED due to failure to meet specifications, either when shipped or due to defects of damage in transit, may be returned to you for credit and are not to be replaced except upon receipt of written instructions from the City.

Inspection. GOODS are subject to the City inspection on arrival. When the words 'no substitute' are used in the article description, only the article described will be acceptable.

Final Acceptance. Receipt of the merchandise, services or equipment in response to this order can result in authorized payment on the part of the City. However, it is to be understood that FINAL ACCEPTANCE is dependent upon completion of all applicable required inspection procedures.

The City is not liable for payment of any good or service shipped or released to the City and County, without the appropriate purchasing authorization.

Freight Terms. Shipments must be F.O.B., City, at the above address, unless otherwise specified on this order. If permission is given to prepay freight and charge separately, the original freight bill must accompany invoice. Additional charges for packing will not be accepted.

Shipment Distance. Where manufacturers have distributing points in various parts of the country, shipment is expected from the nearest distribution point to destination, and excess freight will be deducted from Invoice when shipments are made from greater distance.

Permits. Seller shall procure at sellers sole cost all necessary permits, certificates and licenses required by all applicable laws, regulations, ordinances and rules of the state, municipality, territory or political subdivision where the work is performed, or required by any other duly constituted public authority having jurisdiction over the work of vendor. Seller further agrees to hold the City harmless from and against all liability and loss incurred by them by reason of an asserted or established violation of any such laws, regulations, ordinances, rules and requirements.

Authorization. All parties to this agreement agree that the representatives are, in fact, bona fide and possess full and complete authority to bind said parties.

Any correspondence regarding this purchase order shall be addressed to the Purchasing Division, referencing the purchase order number.

LIMITATION OF TERMS. This Purchase Order expressly limits acceptance to the terms and conditions stated herein set forth and any supplementary or additional terms and conditions annexed hereto or incorporated herein by reference. Any additional or different terms and conditions proposed by seller are objected to and hereby rejected.

2. DELIVERY.
PLEASE ADVISE THE PURCHASING DIVISION or Ordering Department immediately if you cannot make complete shipment to arrive on your promised delivery date as noted. Time is of the essence. Delivery and performance must be effected within the time stated on the purchase order and the documents attached hereto. No acts of the Purchasers including, without limitation, acceptance of partial late deliveries, shall operate as a waiver of this provision. In the event of any delay, the Purchaser shall have, in addition to other legal and equitable remedies, the option of placing this order elsewhere and holding the Seller liable for damages. However, the Seller shall not be liable for damages as a result of delays due to causes not reasonably foreseeable which are beyond its reasonable control and without its fault of negligence, such acts of God, acts of civil or military authorities, governmental priorities, fires, strikes, flood, epidemics, wars or riots provided that notice of the conditions causing such delay is given to the Purchaser within five (5) days of the time when the Seller first received knowledge thereof. In the event of any such delay, the date of delivery shall be extended for the period equal to the time actually lost by reason of the delay.

3. WARRANTY.
The Seller warrants that all goods, articles, materials and work covered by this order will conform with applicable drawings, specifications, samples and/or other descriptions given, will be fit for the purposes intended, and performed with the highest degree of care and competence in accordance with accepted standards for work of a similar nature. The Seller agrees to hold the purchaser harmless from any loss, damage or expense which the Purchaser may suffer or incur on account of the Sellers breach of warranty. The Seller shall replace, repair or make good, without cost to the purchaser, any defects or faults arising within one (1) year or within such longer period of time as may be prescribed by law or by the terms of any applicable warranty provided by the Seller after the date of acceptance of the goods furnished hereunder (acceptance not to be unreasonably delayed), resulting from imperfect or defective work done or materials furnished by the Seller. Acceptance or use of goods by the Purchaser shall not constitute a waiver of any claim under this warranty. Except as otherwise provided in this purchase order, the Sellers liability hereunder shall extend to all damages proximately caused by the breach of any of the foregoing warranties or guarantees, but such liability shall in no event include loss of profits or loss of use.

4. CHANGES IN LEGAL TERMS.
The Purchaser may make changes to legal terms by written change order.

5. CHANGES IN COMMERCIAL TERMS.
The Purchaser may make any changes to the terms, other than legal terms, including additions to or deletions from the quantities originally ordered in the specifications or drawings, by verbal or written change order. If any such change affects the amount due or the time of performance hereunder, an equitable adjustment shall be made.

6. TERMINATIONS.
The Purchaser may at any time by written change order, terminate this agreement as to any or all portions of the goods then not shipped, subject to any equitable adjustment between the parties as to any work or materials then in progress provided that the Purchaser shall not be liable for any claims for anticipated profits on the uncompleted portion of the goods and/or work, for incidental or consequential damages, and that no such adjustment be made in favor of the Seller with respect to any goods which are the Sellers standard stock. No such termination shall relieve the Purchaser or the Seller of any of their obligations as to any goods delivered hereunder.

7. CLAIMS FOR ADJUSTMENT.
Any claim for adjustment must be asserted within thirty (30) days from the date the change or termination is ordered.

8. COMPLIANCE WITH LAW.
The Seller warrants that all goods sold hereunder shall have been produced, sold, delivered and furnished in strict compliance with all applicable laws and regulations to which the goods are subject. The Seller shall execute and deliver such documents as may be required to effect or evidence compliance. All laws and regulations required to be incorporated in agreements of this character are hereby incorporated herein by this reference. The Seller agrees to indemnify and hold the Purchaser harmless from all costs and damages suffered by the Purchaser as a result of the Sellers failure to comply with such law.

9. ASSIGNMENT.
Neither party shall assign, transfer, or convey this order, or any monies due or to become due hereunder without the prior written consent of the other party.

10. TITLE.
The Seller warrants full, clear and unrestricted title to the Purchaser for all equipment, materials, and items furnished in performance of this agreement, free and clear of any and all liens, restrictions, reservations, security interest encumbrances and claims of others.

11. NONWAIVER.
Failure of the Purchaser to insist upon strict performance of the terms and conditions hereof, failure or delay to exercise any rights or remedies provided herein or by law, failure to promptly notify the Seller in the event of a breach, the acceptance of or payment for goods hereunder or approval of the design, shall not release the Seller of any of the warranties or obligations of this purchase order and shall not be deemed a waiver of any right of the purchaser to insist upon strict performance hereof or any of its rights or remedies as to any such goods, regardless of when shipped, received or accepted, as to any prior or subsequent default hereunder, nor shall any purported oral modification or rescission of this purchase order by the Purchaser operate as a waiver of any of the terms hereof.

12. PURCHASERS PERFORMANCE OF SELLERS OBLIGATIONS.
If the Purchaser directs the Seller to correct nonconforming or defective goods by a date to be agreed upon by the Purchaser and the Seller, and the Seller thereafter indicates its inability or unwillingness to comply, the Purchaser may cause the work to be performed by the most expeditious means available to it, and the Seller shall pay all costs associated with such work.

The Seller shall release the Purchaser and its contractors of any tier from all liability and claims of any nature resulting from the performance of such work.

This release shall apply even in the event of fault or negligence of the party released and shall extend to the directors, officers and employees of such party.

The Seller's contractual obligations, including warranty, shall not be deemed to be reduced, in any way, because such work is performed or caused to be performed by the Purchaser.

13. PATENTS.
Whenever the Seller is required to use any design, device, material or process covered by letter, patent, trademark or copyright, the Seller shall indemnify and save harmless the Purchaser from any and all claims for infringement by reason of the use of such patented design, device, material or process in connection with the contract, and shall indemnify the Purchaser for any cost, expense or damage which it may be obliged to pay by reason of such infringement at any time during the prosecution or after the completion of the work. In case said equipment, or any part thereof or the intended use of the goods, is in such suit held to constitute infringement and the use of said equipment or part is enjoined, the Seller shall, at its own expense and at its option, either procure for the Purchaser the right to continue using said equipment or parts, replace the same with substantially equal but non infringing equipment, or modify it so it becomes non infringing.

14. INSOLVENCY.
If the Seller shall become insolvent or bankrupt, make an assignment for the benefit of creditors, appoint a receiver or trustee for any of the Sellers property or business, this order may forthwith be canceled by the Purchaser without liability.

15. GOVERNING LAW.
The definitions of terms used or the interpretation of the agreement and the rights of all parties hereunder shall be construed under and governed by the laws of the State of Utah, USA.

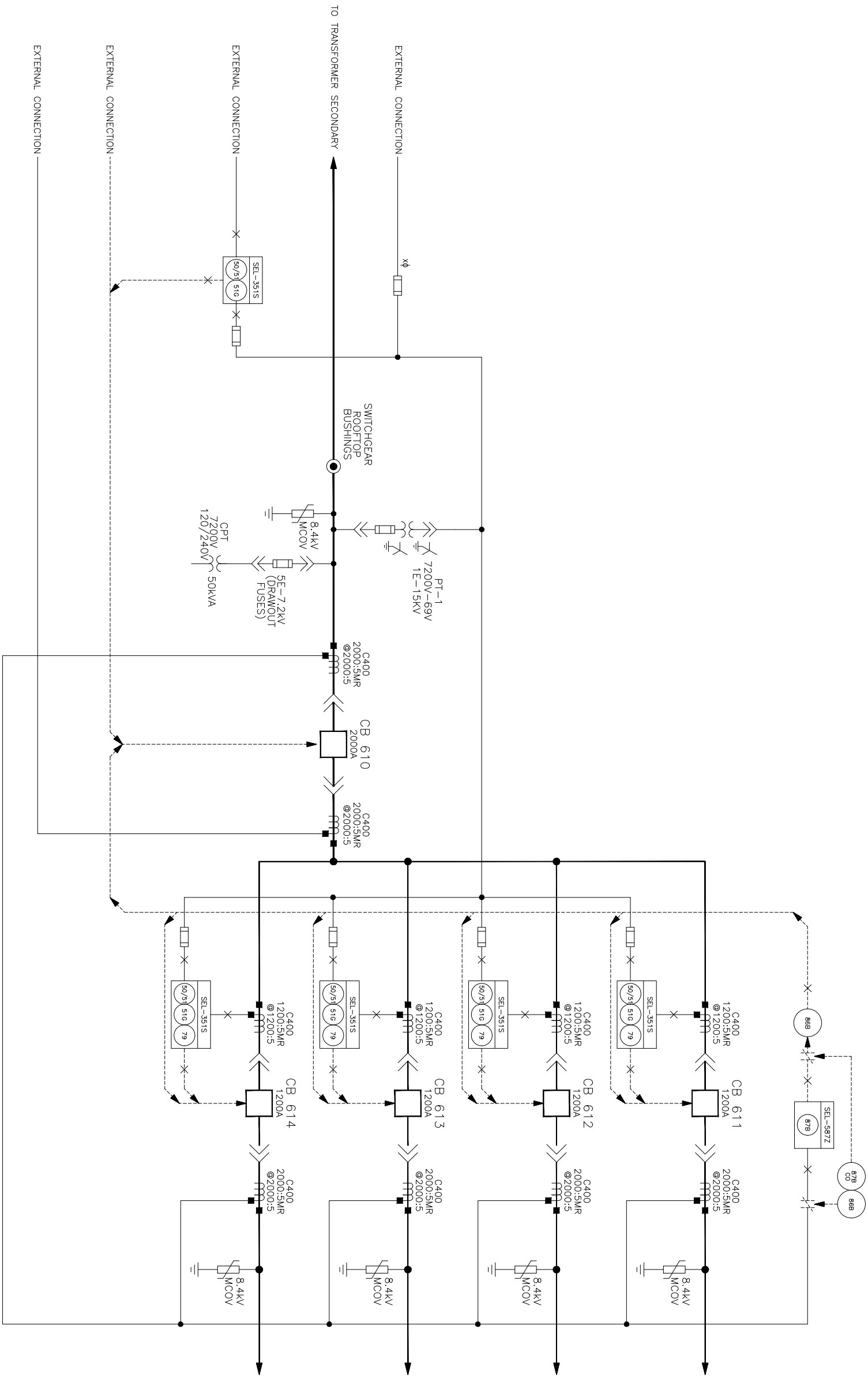
The following Additional Conditions apply only in cases where the Seller is to perform work hereunder, including the services of Sellers Representative(s), on the premises of others.

16. SELLERS RESPONSIBILITY.
The Seller shall carry on said work at Seller's own risk until the same is fully completed and accepted, and shall, in case of any accident, destruction or injury to the work and/or materials before Seller's final completion and acceptance, complete the work at Seller's own expense and to the satisfaction of the Purchaser. When materials and equipment are furnished by others for installation or erection by the Seller, the Seller shall receive, unload, store and handle same at the site and become responsible therefor as though such materials and/or equipment were being furnished by the Seller under the order.

17. INSURANCE.
Unless other wise specified, the Seller shall, at his own expense, provide for the payment of workers compensation, including occupational disease benefits, to its owners and employees employed on or in connection with the work covered by this purchase order, and/or to their dependents in accordance with the laws of the state in which the work is to be done. The Seller shall also carry comprehensive general liability including, but not limited to, contractual and automobile public liability insurance with bodily injury and death limits of at least $300,000 for any one person, $500,000 for any one accident and property damage limit per accident of $400,000. The Seller shall likewise require his contractors, if any, to provide for such compensation and insurance. Before any of the Sellers or his contractors employees shall do any work upon the premises of others, the Seller shall furnish the Purchaser with a certificate that such compensation and insurance have been provided. Such certificates shall specify the date when such compensation and insurance have been provided. Such certificates shall specify the date when such compensation and insurance expires. The Seller agrees that such compensation and insurance shall be maintained until after the entire work is completed and accepted.

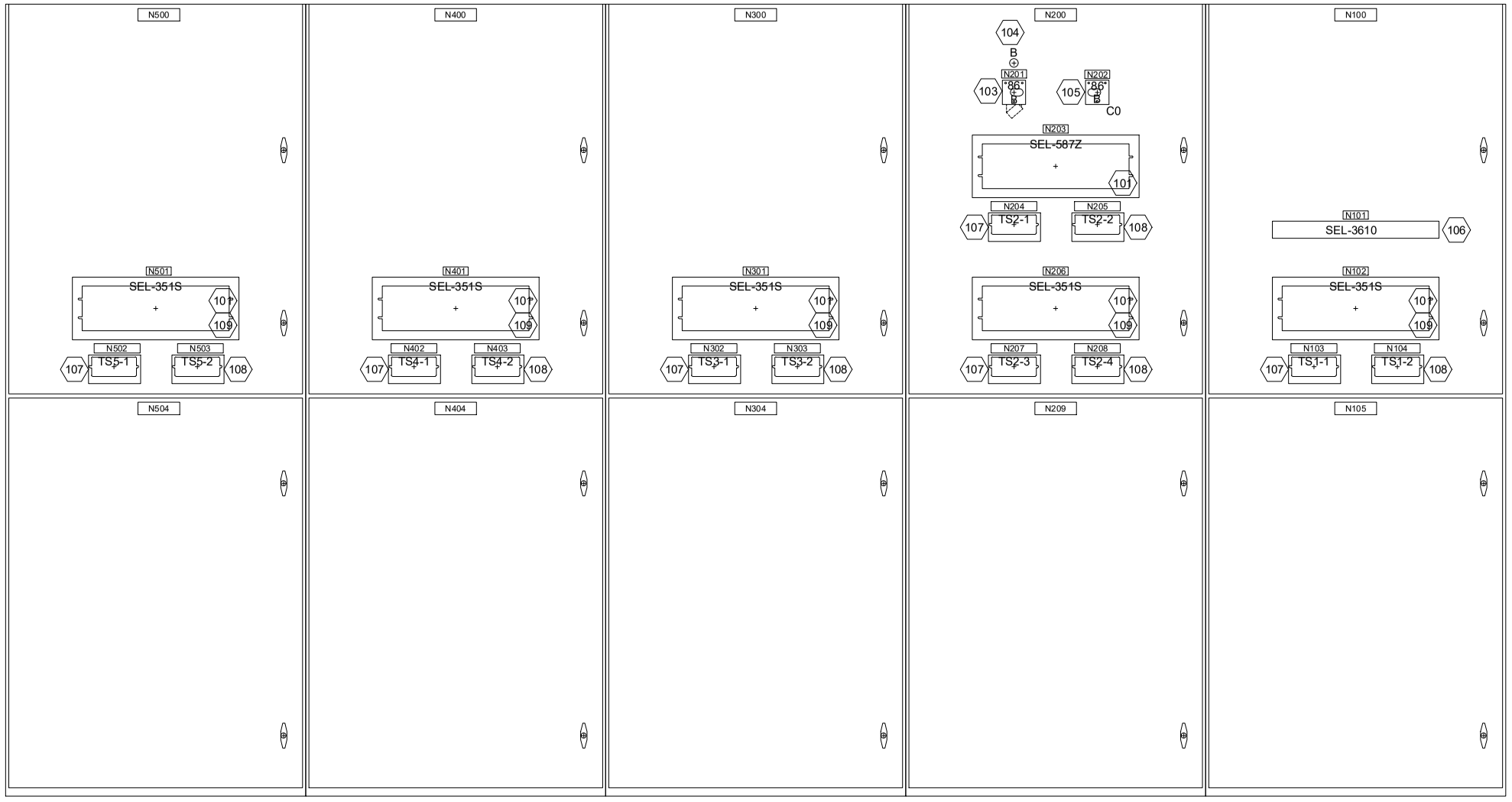
18. PROTECTION AGAINST ACCIDENTS AND DAMAGES.
The Seller hereby assumes the entire responsibility and liability for any and all damage, loss or injury of any kind or nature whatsoever to persons or property caused by or resulting from the execution of the work provided for in this purchase order or in connection herewith. The Seller will indemnify and hold harmless the Purchaser and any or all of the Purchasers officers, agents and employees from and against any and all claims, losses, damages, charges or expenses, whether direct or indirect, and whether to persons or property to which the Purchaser may be put or subject by reason of any act, action, neglect, omission or default on the part of the Seller, any of his contractors, or any of the Sellers or contractors officers, agents or employees. In case any suit or other proceedings shall be brought against the Purchaser, or its officers, agents or employees at any time on account or by reason of any act, action, neglect, omission or default of the Seller of any of his contractors or any of its or their officers, agents or employees as aforesaid, the Seller hereby agrees to assume the defense thereof and to defend the same at the Sellers own expense, to pay any and all costs, charges, attorneys fees and other expenses, any and all judgments that may be incurred by or obtained against the Purchaser or any of its or their officers, agents or employees in such suits or other proceedings, and in case judgment or other lien be placed upon or obtained against the property of the Purchaser, or said parties in or as a result of such suits or other proceedings, the Seller will at once cause the same to be dissolved and discharged by giving bond or otherwise. The Seller and his contractors shall take all safety precautions, furnish and install all guards necessary for the prevention of accidents, comply with all laws and regulations with regard to safety including, but without limitation, the Occupational Safety and Health Act of 1970 and all rules and regulations issued pursuant thereto.

19. CONFLICTS WITHIN THE CONTRACT DOCUMENTS: In the event that conflicts exist within the above stated purchase order terms and conditions and the attached and/or referenced contract documents the latter shall supersede.



CE-002	SUB 6	DRAWN BY:	CWM			NO.	DATE:	BY:	REVISION DESCRIPTION:	NO.	DATE:	BY:	REVISION DESCRIPTION:
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ATTACHMENT 6.3



UNIT 5
SPACE
6-14

UNIT 4
SPACE
6-13

UNIT 3
STATION SERVICE
6-12

UNIT 2
MAIN BUS CONNECTION POINT
6-11

UNIT 1
PT-1
6-0

RELAY LIST

ITEM	DEVICE	STYLE NUMBER	QTY	MFR	DESCRIPTION
101	351S	0351S5X3D3E52X1	5	SEL	RELAY PACKAGE, SEL-351S 48VDC POWER SUPPLY AND CONTROL, 5AMP PHASE AND NEUTRAL SECONDARY, STD. TERMINAL BLOCKS, RS-232 PORTS, 3 PHASE VOLTAGE AND SYNCH
102	587Z	0587Z0X225312XX	1	SEL	RELAY, TYPE SEL-587Z HIGH-IMPEDANCE DIFFERENTIAL RELAY 5A PHASE SECONDARY, 48 VDC CONTROL VOLTAGE, 2 MOVS 5000J, 2000V HORIZONTAL PANEL MOUNT, 48/125 VDC OR VAC POWER SUPPLY.
103	86-B1	7808C	1	ELECT/SW	LOCKOUT RELAY, SERIES 24 LOR, 8 DECK, 48 VDC
104	B	116B6708G2-B	1	G.E.	INDICATING LIGHT LED, TYPE ET-16, 48 VDC, WITH RESISTOR AND BLUE LENS.
105	87T1-C0	24204B	1	ELECT/SW	CUTOUT SWITCH, SERIES 24 4 DECKS, SINGLE THROW "NORMAL" TEXT AT 12 O'CLOCK "CUTOUT" TEXT AT 2 O'CLOCK
106	3610	3610X3B1XXX0	1	SEL	PORT SERVER - 17 PORTS,48VDC, ONE 10/100BASE-T, ONE 100BASE-FX (MULTIMODE)
107	TS1-1 TS2-1 TS2-3 TS3-1 TS4-1 TS5-1	C129A514G01	6	ABB	TYPE FT-1 2RU, CLEAR P P P C-C C-C C-C P
108	TS1-2 TS2-2 TS2-4 TS3-2 TS4-2 TS5-2	C129A539G01	6	ABB	TYPE FT-1 2RU, CLEAR T T T T T T T T T T
109	SERIAL-1 SERIAL-2 SERIAL-3 SERIAL-4 SERIAL-5 SERIAL-6	C273A-** ** LENGTHS TBD	6	SEL	SERIAL CABLES FROM ALL RELAYS TO SEL-3610
110	FX	H25030-3S	AS REQ	BUSSMAN	FUSE BLOCK, 3-POLE, 30AMP, 250 VOLT DEVICE.
111	FUSE	NON5	AS REQ	BUSSMAN	FUSE, TYPE NON, ONETIME, CLASS H, 5A.
112	TB1	EB25B12	AS REQ	G.E.	TERMINAL BOARD, 12 POLE, 600 VOLT, 30 AMP.
113	TB2	EB27B04S	AS REQ	G.E.	TERMINAL BOARD, EB-27, 4-CIRCUIT SHORTING TYPE, 600 VOLT, 30 AMP.

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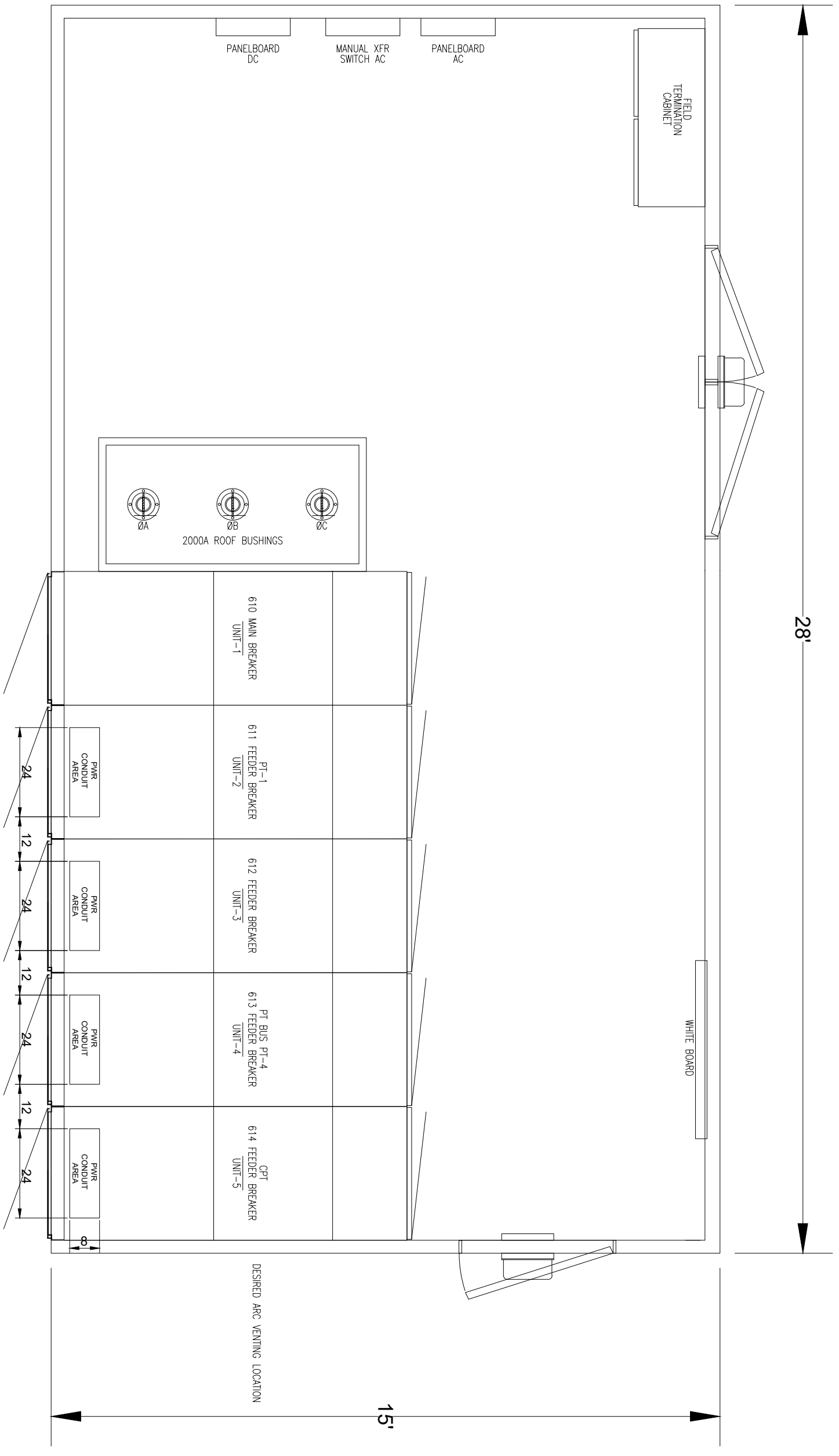
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SUB 6

SWGR ELEVATION

C1-002



PLAN VIEW

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		SHEET NUM:	1 SHEET SIZE: D		3				8			
	SWGR LAYOUT	LOCATION:			4				9			
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